

Seasonal water relations of *Lyginia barbata* (Southern rush) in relation to root xylem development and summer dormancy of root apices

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Summary

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Key words: cryo-energy dispersive X-ray microanalysis, cryo-scanning electron microscopy, dormant roots, drought resistance, monocotyledon, Restionaceae, xylem differentiation, xylem–phloem juxtaposition.

- Periods of dormancy in shallow roots allow perennial monocotyledons to establish deep root systems, but we know little about patterns of xylem maturation, water-transport capacities and associated economies in water use of growing and dormant roots.
- Xylem development, anatomy, conductance and *in situ* cellular [K] and [Cl] were investigated in roots of field-grown *Lyginia barbata* (Restionaceae) in Mediterranean southwestern Australia. Parallel studies of gas exchange, culm relative water loss and soil water content were conducted.
- Stomatal conductance and photosynthesis decreased during summer drought as soil profiles dried, but rates recovered when dormant roots became active with the onset of wetter conditions. Anatomical studies identified sites of close juxtaposition of phloem and xylem in dormant and growing roots. Ion data and dye tracing showed mature late metaxylem of growing roots was located ≥ 100 mm from the tip, but at only ≤ 10 mm for dormant roots. Dormant roots remained hydrated in dry soils (0.001–0.005 g g⁻¹).
- Effective regulation of growth and water-conserving/obtaining properties permits the survival of shallow roots of *L. barbata* during summer drought and may represent important strategies for establishing deeper perennial root systems in other monocotyledonous plants adapted to seasonally dry habitats.

Abbreviations: CEDX, cryo energy dispersive X-ray microanalysis; CRWL, culm relative water loss; CSEM, cryo-scanning electron microscopy; EMX, early metaxylem; LMX, late metaxylem; WCS, water content of soil.

Introduction

In the late 1960s, McWilliam & Kramer recognised that during drought, survival of the Mediterranean grass *Phalaris tuberosa* ‘appears to depend largely on the plant’s ability to maintain an adequate water supply at the bases of the dormant culms’. Their results established that those dormant culms and shallow, dormant roots were sustained by water extracted by perennial roots *c.* 2 m deep in the soil profile (McWilliam, 1968; McWilliam & Kramer, 1968). These results implied that the perennial ‘deep’ roots of

grasses, and presumably other perennial monocotyledons, are vital determinants of survival during prolonged summer drought in Mediterranean-type environments. However, the water-conserving/obtaining properties permitting the survival of those younger, shallowly located and seasonally dormant perennial roots were never further characterized by these authors.

Perennial grasses coping with harsh temperate environments deploy a range of above-ground strategies to survive extreme drought. Usual responses include reduced shoot growth, shedding of mature foliage and/or formation of

resting organs, but survival of mature plants ultimately depends upon established and functional perennial 'deep' root systems (reviewed in Voltaire & Norton, 2006). More exceptional responses occur with perennial 'resurrection' grasses in Australia and South Africa, whose shoots survive by becoming severely desiccated and then fully restoring their water content when wet conditions return (Gaff & Latz, 1978; Gaff, 1981). However, almost nothing is known about comparable survival strategies in the roots of such species. Interestingly, Schneider *et al.* (1999) showed that a lipid film 'internal cuticle' partly covers inner walls of xylem vessels during the 'dry' state in shoot branches of the 'woody' resurrection plant *Myrothamnus flabellifolia*, potentially restricting loss of water from adjacent xylem parenchyma cells. Root-soil sheaths and root-soil-air gaps that limit root loss and maintain higher water potentials within root tissues compared with that of dry soil are also known for succulent plants adapted to desert environments (Nobel & Cui, 1992; Huang *et al.*, 1993; North & Nobel, 1997).

We have been investigating survival strategies of perennial roots in a common and widespread Southern rush, *Lyginia barbata* (Restionaceae). It is endemic to the sandplain environment of Mediterranean southwestern Australia (Briggs & Johnson, 1999) where mild, wet winters favour plant water and nutrient uptake, and hot, dry summers challenge their survival. This rush species is long-lived, with additional culms and perennial roots developing on its elongating rhizomes each winter. In mature plants, the oldest perennial roots were traced to a depth of approx. 2 m in the nutrient-deficient sandplain habitat (Fig. 1) but Pate & Meney (1999) have shown that some roots may penetrate as deeply as 4 m. These 'deep' roots, as with those of the perennial grasses described earlier, presumably supply water to the above-ground culms which, in *Lyginia*, are 200–300 mm high and typically remain green and hydrated during summer drought. In this scenario, nothing is known about the net assimilation rates of Restionaceae, how these vary seasonally and how carbon acquisition relates to patterns of transpiration and water use-efficiency (Meney & Pate, 1999).

Recently, we reported that root dormancy (i.e. ‘a state of inactivity imposed by environmental conditions unfavourable to growth rather than by physiological state’ – Sutton & Tinus, 1983) in *L. barbata* during summer drought is critical for survival of younger perennial roots at early stages of their downward path through the sandy rooting substrate of this habitat (Figs 1,2). Owing to slow growth rates (*c.* 0.5 m per growing season) and the timing of initiation, a root may survive for at least two summers in dry, hot soil before reaching deep soil moisture (Shane *et al.*, 2009). Periods of root dormancy during their downward passage allow these ‘shallow’, perennial roots to survive each summer to the following winter, when root elongation resumes. This cycle will typically be repeated

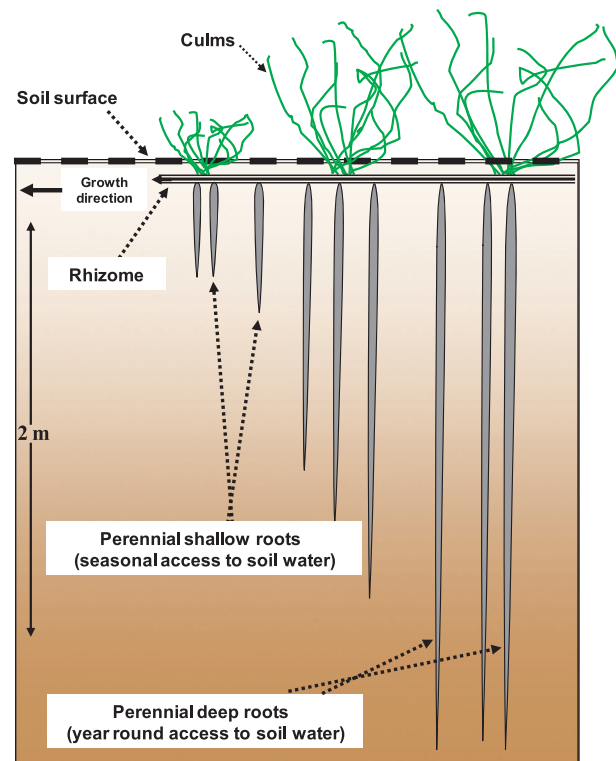


Fig. 1 Simplified scheme of mature *Lyginia barbata* root, rhizome and shoot system. Culms are 1–2 mm in diameter and 20–550 mm long and can be photosynthetically active year-round. Perennial deep roots penetrate 2–4 m into the soil profile and can function in water uptake all year. Younger, perennial shallow roots grow in winter and are summer-dormant in dry soil. A widely looping 100–250 mm deep rhizome links perennial roots to culms.

over several years until roots eventually extend 2–4 m deep (Fig. 1). Summer-dormant roots remain relatively well hydrated (*c.* 70% water content) even though they survive the summer in bone-dry soil, well out of reach of water (Shane *et al.*, 2009).

Rhizosheaths of sand grains trapped tightly by long, tangled, persistent root hairs surround the tips of dormant roots, often completely encasing them (Fig. 2b,c), whereas elongating roots are bare for up to 15 mm from their apices (Fig. 2d,e). Thick sand sheaths and osmotic adjustment partly account for the relatively good hydration state of summer-dormant roots, but nothing is known about vascular water-conserving/obtaining properties that permit the survival of still shallow roots (Shane *et al.*, 2009).

Our previous findings with *Lyginia* have raised questions about whether patterns of xylem maturation, water-transport capacities and associated economies in water use vary between the dormant and growing shallow roots and how these may contribute to their survival in such a harsh environment. This study explores xylem development in these shallow roots using dye tracing of water flow and cryo-analytical scanning electron microscopy. Linking these data

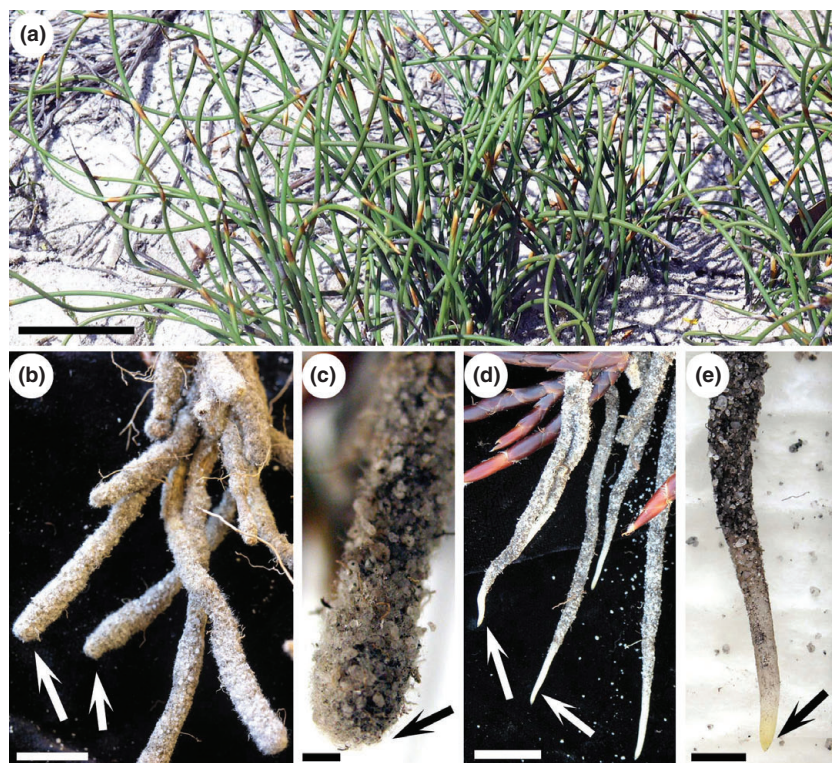


Fig. 2 (a) 'Grasslike' culms of *Lyginia barbata* in winter. (b, c) Summer-dormant perennial roots. Tips (arrows) are encased by sand sheath. (d, e) Winter-growing perennial roots. Tips (arrows) are free of sand sheath. Bars: a, 28 mm; b, 9 mm; c, 1.6 mm; d, 16 mm; e, 4 mm.

with monthly measurements of culm gas exchange, determinations of culm relative water loss and water content in the soil, we are then able to construct a general description of the water relations of *L. barbata*.

Materials and Methods

Plant material and growth conditions

Summer-dormant and winter-growing shallow perennial roots of *Lyginia barbata* R.Br. (Restionaceae, Southern rushes) were sampled on southwestern Australian sandplains over 3 yr (2006–2008) on a population of > 100 plants at Melaleuca park (50 km north of Perth) and Yule Brook reserve (25 km southeast of Perth). *Lyginia* species are typically adapted to nutrient-impovertised sands of the Bassendean Dune System (McArthur, 1991) and other low-nutrient substrates (Meney & Pate, 1999) where they regularly experience several months of heat and limited rainfall from summer to early autumn. The climate is Mediterranean with cool, wet winters (growing roots collected June–September) and hot, dry summers (dormant roots collected November–February). For additional information on the field site, see Shane *et al.* (2009).

Gas exchange and culm relative water loss

Monthly or bimonthly gas-exchange measurements were taken between 10:00 and 12:00 h on the youngest fully

matured culms (six culms per plant, 1–2 yr old) from each of five *L. barbata* plants growing in the field using a Li-6400 Portable Photosynthesis System (Li-Cor, Lincoln, NE, USA), PAR 1500 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$ and vapour pressure deficit of air supplied to the cuvette 1.5–5.0 kPa depending upon the season. Photosynthesis rates and stomatal conductance were calculated using the Li-6400 software. After the gas-exchange measurements, culms were excised and sealed in airtight plastic vials, kept on ice, and returned to the laboratory for measurement of relative water loss. Relative water loss was assessed according to Canny & Huang (2006) as $100(W_s - W_f)/W_s$, where W_f is the weight of fresh matter at harvest and W_s is the saturated weight measured after rehydration of the culm at 4°C for 48 h in the dark. Relative water loss (RWL) can be converted to relative water content (RWC) using $\text{RWC} = 1 - \text{RWL}/Q_x$ (Canny & Huang, 2006) where Q_x is the maximum mass fraction of water. For *Lyginia* culms, $Q_x = 49\%$ (SE 0.5); thus a RWL of 4% corresponds to $\text{RWC} = 92\%$.

Rooting depths, soil and root water status

Excavations were made to determine the depth of root penetration and the water content of soils collected at three depths (i.e. 70–100 mm, 500–570 mm and 1500–1570 mm; three replicates within 100 m²). Soils were first weighed fresh (FW) and then after drying at 70°C for 3 d (DW). The water content of soil (WCS) was calculated as

$WCS = (FW - DW)/DW$. Growing or dormant perennial roots were sampled in the top 1 m of soil. The water content of these roots was reported in an earlier paper (Shane *et al.*, 2009). Briefly, the method determined the fresh weights of root samples collected in the field (with sand sheaths removed) and dry weights after drying roots to constant weight at 70°C. The percentage water content (mass fraction of water) was calculated as $100(FW - DW)/FW$. These measurements are summarized and used in the present study, but cannot be converted to RWL for comparison with culms, in the absence of data for W_s .

Root collection in the field for microscopy and analysis

For cryo-microscopy, whole lengths of winter-growing and summer-dormant roots were carefully excavated around *Lyginia* plants growing in the sandplain habitat. Roots were excised and immediately frozen intact by plunging into liquid nitrogen (LN₂). Frozen roots were stored and transported to the laboratory in a cryo-shipper and stored in LN₂ until shipped in a cryo-shipper to Canberra for microscopy and analysis.

For optical microscopy and dye tracer studies, whole lengths of winter-growing and summer-dormant roots were freshly excavated and excised in the field as described, wrapped in damp paper towel and transported to the laboratory in plastic bags in a cool box at *c.* 4°C.

Root structural studies

Sectioned material for optical microscopy Tissues, fresh or fixed in 10% (v/v) formalin, were sectioned by hand and observed by bright-field and fluorescence optics, unstained or after staining with 0.05% (w/v) toluidine blue (pH 4.4) or after mild vacuum infiltration of basic fuchsin to identify currently conducting vessels. To identify the phloem, other sections were stained for 60 min in rhodamine B (10 ppm, w/v), washed for 5 min in tap water and then mounted in aniline blue solution (0.05% aqueous w/v, pH 8.5). Bright-field and epifluorescence optics (filters, UV exciter, 'blue' barrier or UV exciter, 'violet' barrier) were with a Zeiss Axiophot (Oberkochen, Germany). Photomicrographs were recorded in colour on Agfa RS100 colour slide film, scanned into digital format (Nikon, Coolsan 5000, Tokyo, Japan) and processed using Photoshop CS2 software (San Jose, CA, USA).

Cryo-fixed tissues examined fully hydrated by cryo-scanning electron microscopy (CSEM) Frozen pieces (*c.* 1 cm) were cut from the desired regions (specific distances proximal to root apices) under LN₂, placed in cryo-vials and held in a cryo-store at -196°C until examined. The sand sheath was scraped off the root surfaces while still frozen. A segment (*c.* 4 mm long) was cut from the centre of each frozen

root piece under LN₂ and quickly secured in the groove of an aluminium stub with low-temperature tissue mountant and plunged immediately into LN₂. The stub with frozen tissue was then transferred under LN₂ to the chuck of the cryo-microtome (CR 2000, Research and Manufacturing, Inc., Tucson, AZ, USA). A transverse or longitudinal face was planed flat with glass and diamond knives at -80°C. The specimen was then transferred with a cryo-transfer system (CT 1500; Oxford Instruments, Eynsham, Oxford, UK) to the stage of a CSEM (Hitachi 4300 Schottky Field Emission, Tokyo, Japan) and lightly etched at -90°C to reveal the outlines of the cells. It was then re-cooled to -170°C, coated with evaporated Al, and examined at 7–15 kV. Details of these procedures are given in Huang *et al.* (1994) and McCully *et al.* (2000).

Vessel numbers and diameters Measurements of vessel diameters were made on digitally enlarged images of hand-sections (stained with basic fuchsin; see next section) of the roots and the frozen preparations in the CSEM. Wherever sectioned vessels were not of circular conformation, their largest diameter was recorded.

Root vessel maturation

Tracer dye moved under vacuum Dye tracer measurements were made within 4 h of sampling roots in the field. For the growing roots, the most distal segment (0.5 cm), including the tip, was removed and not included in dye tracing measurements. Roots were cut into segments 10–30 mm under water. Basic fuchsin (1% w/v, freshly filtered using a 0.45 µm syringe filter) was drawn basipetally through the vessels of each segment under partial vacuum (*c.* 20 kPa), using a hand-held vacuum pump (Shane *et al.*, 2000). The distal end of the segment was immersed in dye solution. At no time did the root segment dry appreciably. Pressure was reduced until tracer solution had been pulled through the vessels, or after 15 min for those root pieces that held the vacuum. The segment was then transferred to water, and the lowered pressure reapplied until no more dye was present in the liquid that had been drawn through the xylem. Vessels through which dye had passed were then distinguished by transverse sectioning.

Cryo-energy dispersive X-ray microanalysis (CEDX) To distinguish still living vessel elements from mature functioning vessels, their contents, and those of the vacuoles of adjacent xylem parenchyma cells, were analysed for concentrations of K and Cl by CEDX. Based on previous experimental studies using the same CEDX method (Huang *et al.*, 1994; Enns *et al.*, 1998; McCully *et al.*, 2000), we have chosen the arbitrary value of 25 mM for [K] above which a vessel element is likely to be alive and not yet transporting. Also, we assume that there is no statistically significant

cant difference between values for K or Cl concentrations below 25 mM with the technique used (C. X. Huang, unpublished).

Planned transverse faces of frozen root pieces were prepared and examined as described earlier. Microanalysis of vacuole contents of individual cells was carried out with an INCA Energy 450 EDXA system (Oxford Instruments, Oxfordshire, UK), using the Be-window. The accelerating voltage was 15 kV, take-off angle 33°, working distance 35 mm, and probe current set at 0.300 nA (measured using a Faraday cup). Magnification was varied so that the raster covered the lumen of the cells but did not touch the wall. Spectra were collected for a fixed live time of 100 s. Potassium and Cl concentrations were determined by comparison with the appropriate standards frozen, cryo-planed and analysed identically (Huang *et al.*, 1994; McCully *et al.*, 2000).

Results

Plant water status and activity with changing seasons

Culm relative water loss (CRWL) Values for CRWL (Fig. 3a) were between 3.5 ± 0.3 and $4.0 \pm 0.2\%$ during the relatively unstressed conditions of winter–spring, and then increased to between 5 ± 0.5 and $6 \pm 0.4\%$ during the summer–autumn.

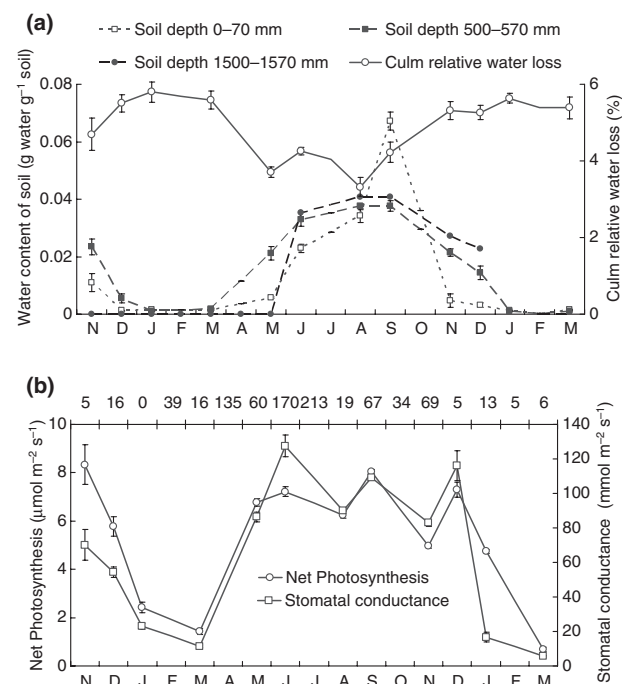


Fig. 3 Monthly variation in culm relative water loss and water content of soil at three depths (a), and photosynthesis and stomatal conductance of *Lyginia barbata* (b). Values across upper x-axis represent rainfall (mm) on a monthly basis during the study (lower x-axis).

Water content of soil (WCS) Values of WCS (Fig. 3a) ($\pm$ SE) measured in May to October (winter–spring ‘wet’ months) ranged from 0.02 ± 0.006 to 0.07 ± 0.003 g g⁻¹ for topsoils (i.e. 0–70 mm depth), and from 0.02 ± 0.002 g g⁻¹ to 0.04 ± 0.001 g g⁻¹ for soil 500–570 and 1500–1570 mm in the profile. From December to March (summer–autumn ‘dry’ months) values were low to extremely low, ranging from 0.001 ± 0.0002 to 0.005 ± 0.0002 g g⁻¹ in topsoils, and from 0.0018 ± 0.0003 to 0.0054 ± 0.0015 g g⁻¹ deeper in the profile. We did not sample 1500–1570 mm in the profile during dry months.

Photosynthesis and stomatal conductance (Fig. 3b) Photosynthesis and stomatal conductance followed seasonal changes in CRWL and water content of soil fairly closely (see later). Thus, when CRWL was low, the mean photosynthesis was 7 ± 3 μmol m⁻² s⁻¹ (maximum 7, minimum 6, $n = 100$) and conductance was 105 ± 7 mmol m⁻² s⁻¹ (maximum 128, minimum 86, $n = 100$), whereas at times when the CRWL was close to maximum, photosynthesis was 4 ± 1 μmol m⁻² s⁻¹ (maximum 5, minimum 1, $n = 200$) and conductances decreased to as low as 6 mmol m⁻² s⁻¹ (mean 41 ± 12 , maximum 116, $n = 200$).

Water in roots The water content of winter-growing roots was $85 \pm 1\%$, and for summer-dormant roots was $59 \pm 1\%$ (Shane *et al.*, 2009).

Root growth All roots with tips located in the top 1 m of the soil surface remained dormant during the dry months from December to March (Fig. 2b,c; for additional detail, see Shane *et al.*, 2009), whereas in wet months, between May and September, roots similarly located were elongating (Fig. 2d,e).

Root anatomy and vessel numbers and diameters

Winter-growing roots The basal regions of roots had 74 ± 3.7 (SE) early metaxylem (EMX) vessels of near-constant diameter (12 ± 0.5 μm, range 8.3–14 μm, $n = 12$) (Figs 4, 5) and 28 ± 2.1 late metaxylem (LMX) vessels. The diameters of the latter vessels (12 measured from nine roots) were distributed more or less normally around a mean of 47 ± 1.9 μm, with a range of 40.6–59.5 μm.

Hand-cut sections (Fig. 4a,b) illustrate the basic features of the stele and endodermis of a typical root. Intervening parenchyma separated the EMX from the LMX and the pericycle from the outermost EMX and phloem (Fig. 4b). Phloem poles include varying numbers of cells, the walls of which stain pink/purple with toluidine blue, in contrast to the blue/green stain shown by all other walls (Fig. 4a). The phloem cell walls are further characterized by their bluish fluorescence following rhodamine B/aniline blue staining (Fig. 4b). Frequently (Figs 4a–c, 6), there was no interven-

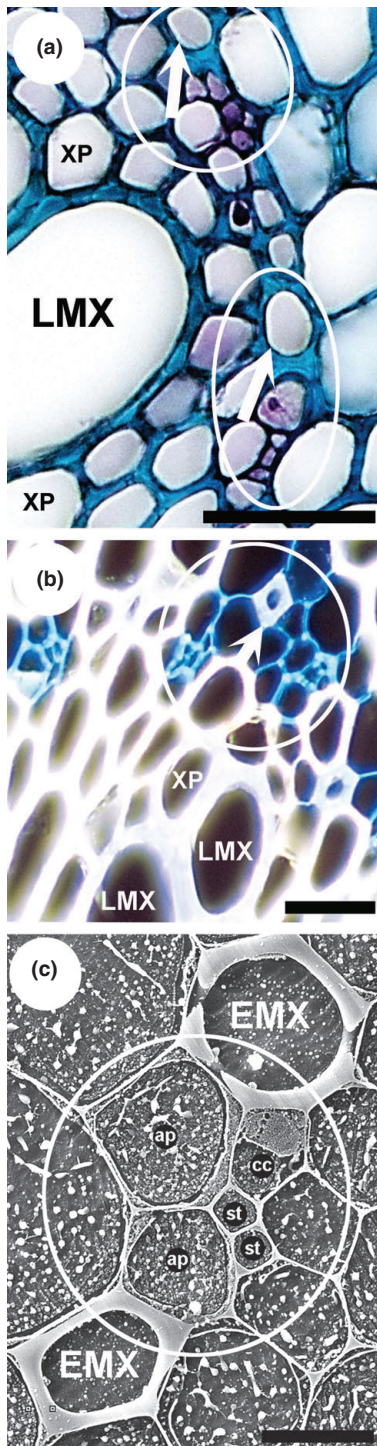


Fig. 4 (a, b) Transverse, hand-cut sections of winter-growing shallow roots of *Lyginia barbata* sampled ≤ 1 m from the soil surface. Phloem poles adjacent to early metaxylem (EMX) vessels (arrows) are circled. Phloem cell walls are thin, while those of both EMX and late metaxylem (LMX) vessels and xylem parenchyma (XP) are much thickened and secondary. (a) Toluidine blue stain (bar, 60 μ m). (b) Rhodamine B and aniline blue-induced fluorescence (bar, 65 μ m). (c) Cryo-planned face of a growing root 10 mm from the tip, frozen in the field (cryo-scanning electron microscopy). The phloem complex (circled) adjoins both EMX vessels (bar, 15 μ m). cc, companion cell; st, sieve tube; ap, associated phloem cells.

diameter had not yet attained their mature size and shape (Fig. 5b). At 50–100 mm from the apex, many cortical cells had collapsed, and endodermis, LMX and xylem-parenchyma had developed thick secondary walls (Fig. 5c,d). In the oldest, more basal regions analysed (180–200 mm from the apex), all stelar cells (except some phloem cells) and the endodermis possessed extremely thick secondary walls (Fig. 5e,f).

Xylem-vessel elements (EMX and LMX) and xylem parenchyma were readily distinguishable in tissues planed for cryo-analysis at 10, 50–100 and 180–200 mm from apices (Figs 4, 5). Those planed surfaces of the contents of living cells which included immature vessel elements demonstrated high electron emissivity from solutes sequestered in their vacuoles (Figs 4,5). The eutectics patterns (i.e. phase separation that occurs when a mixture of solutes and liquids are frozen; McCully *et al.*, 2009) that were clear in immature vessel elements were absent from mature vessels in growing roots. Immature (not yet transporting) vessel elements, with thin peripheral cytoplasm, tonoplasts (e.g. LMX, Fig. 5b,d) and intact primary end walls of vessel elements were best visualized in longitudinal faces (Fig. 5g), whereas frozen xylem liquid in mature (transporting) LMX and EMX vessels appeared black, and therefore of low electron emissivity and solute content (Fig. 5f). Mature vessel elements with thick secondary walls and no remaining primary end walls of their component vessel elements showed low solute content. Such vessels were linked in series by lignified, angled vessel end walls with numerous pits that were best visualized in longitudinal faces (Fig. 5h).

Summer-dormant roots Dormant roots were quite different from growing roots in possessing no regions similar to those in Fig. 5a–d. Dormant root tissues were fully differentiated close to (1–3 mm behind) the apex. Here, cells of the stele had already matured and had thick secondary walls (except some phloem cells) (Fig. 6).

Water pathways revealed by dye tracing

Winter-growing roots Dye solution drawn through short (10–30 mm) decorticated segments (Fig. 7a–f) showed

ing parenchyma, and sieve tubes or their companion cells directly adjoined EMX vessels. Most of the central portion of the root comprised thick-walled (lignified) pith parenchyma.

Young regions of growing roots (*c.* 10 mm from the apex) showed an epidermis, 10–15 layers of intact cortical cells, and all its tissues, except the EMX, were thin-walled (Fig. 5a,b). The still-developing LMX elements of large

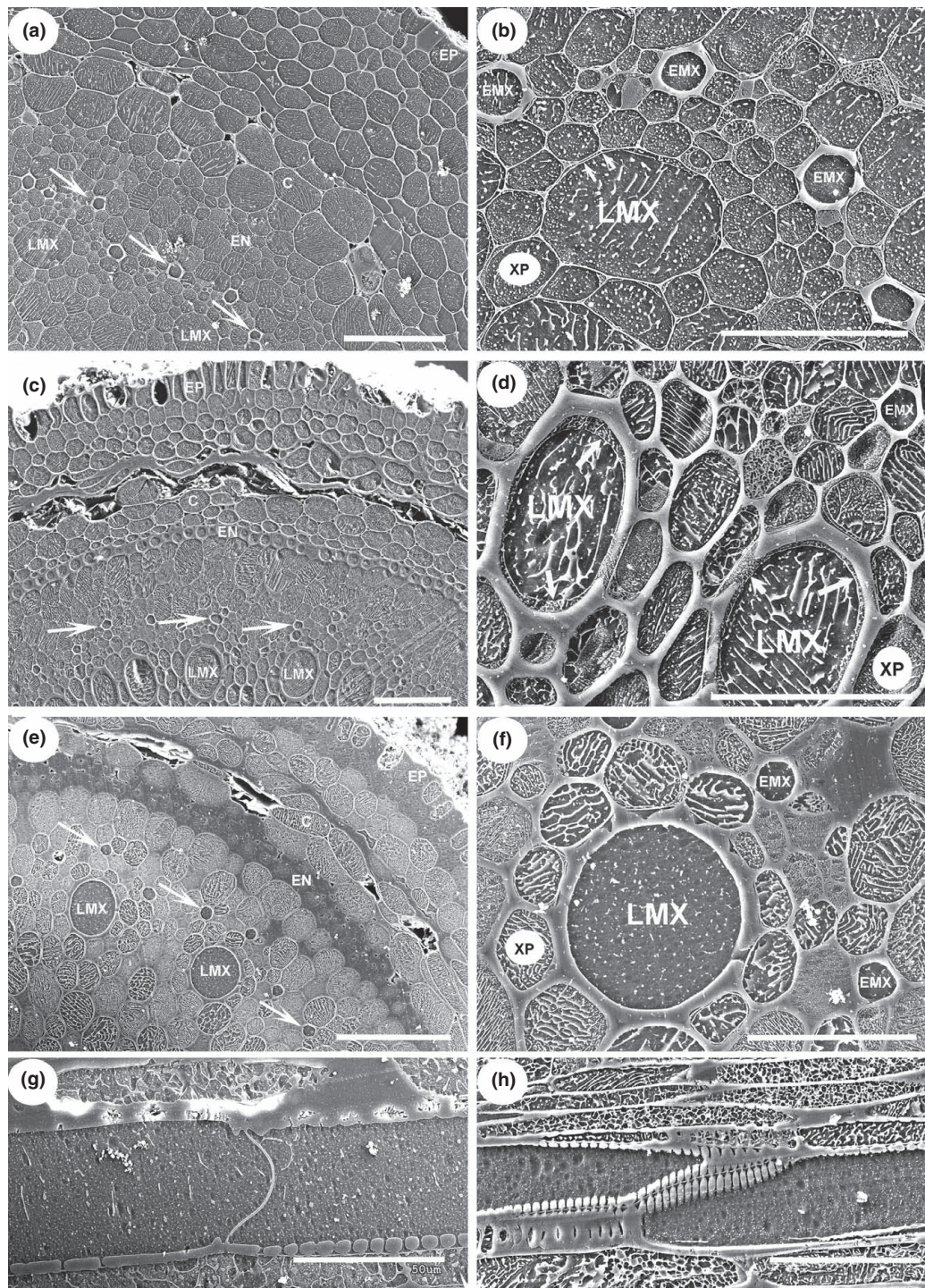


Fig. 5 Cryo-planned faces of winter-growing roots of *Lyginia barbata* frozen in the field (cryo-scanning electron microscopy). (a–f) Cryo-planned transverse faces; (g, h) longitudinal faces – 10 mm (a, b), 50–100 mm (c, d) and 180–200 mm (e, f) from the tip. Arrows mark early metaxylem (EMX) vessels in (a), (c) and (e). At 10 mm from the tip, all cell walls, except those of the EMX vessels, are thin and the cortex is intact. The late metaxylem (LMX) vessel elements are still thin-walled, with intact tonoplasts (arrows) (b). At increasing distance from the tip, the cortex progressively disintegrates (c, e) and cell walls thicken, particularly those of the endodermis (c, e) and developing LMX vessels (d, f). (c, d) LMX vessel elements are still alive, as indicated by the prominent eutectics in their vacuoles (see text) and still intact tonoplasts (arrows) (d). The LMX vessels in (e) and (f) are mature. (g) Region similar to that in (c) and (d) where the lateral walls of the developing LMX vessel are thickened, solutes have been lost from the vessel element vacuoles (no eutectics) but the primary wall between the elements is still intact. (h) Mature region showing an end wall between two mature LMX vessels. C, cortex; EN, endodermis; EP, epidermis; XP, xylem parenchyma. Bars, 100 μ m (a, c, e); 50 μ m (b, d, f, g); 80 μ m (h).

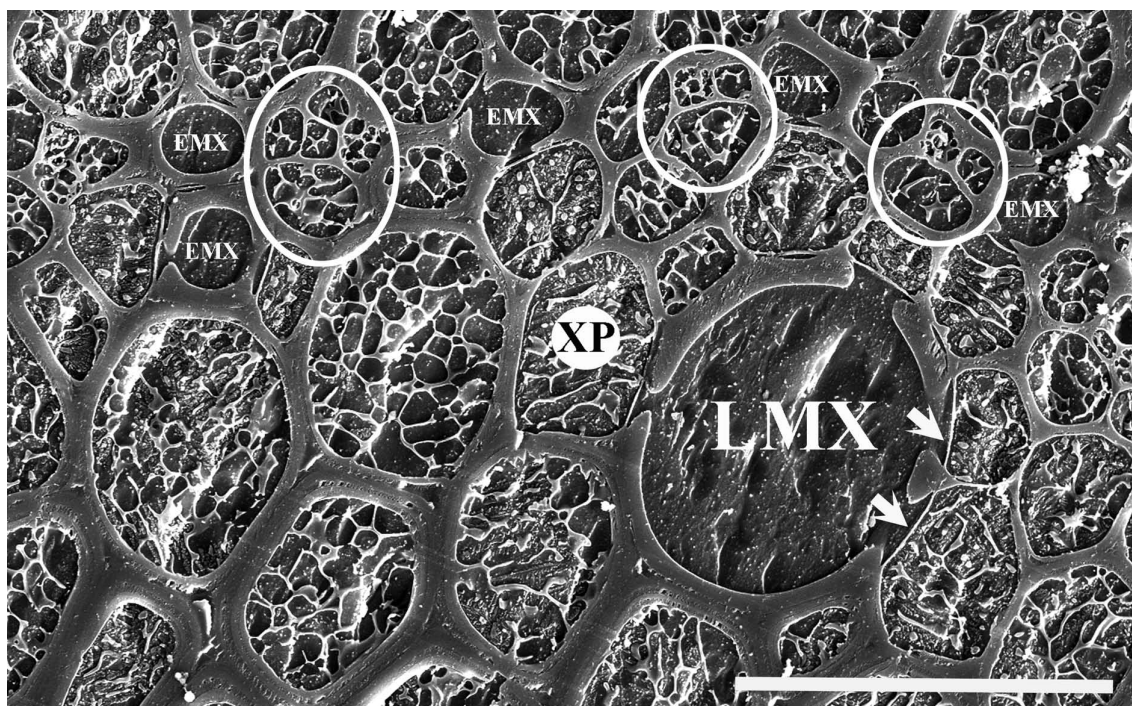


Fig. 6 Cryo-planned, transverse face of dormant root of *Lyginia barbata* 1–3 mm from the tip frozen in the field and viewed under the cryo-scanning electron microscope. All cells except those of the phloem have thick secondary walls, as do the late metaxylem (LMX) and early metaxylem (EMX) vessels. The number of cells of the phloem poles are variable, but in many places (circled) these poles are directly adjacent to EMX vessels, while they are always separated from the LMX vessels by parenchyma cells. Pit membranes between xylem parenchyma (XP) and LMX are indicated by arrows. Bar, 55 μ m.

that over 50% of the EMX of roots were stained back to 10–50 mm from the root apex (Fig. 7a,b). By 100–160 mm, all the EMX vessels were stained, as was also observed at 200–300 mm (Fig. 7e,f), the oldest region of root tested. By contrast, dye solution was not conducted and vessels were accordingly not stained in LMX sited 10–50 mm from the root tip (Fig. 7a,b). At 100–160 mm, only 20% of the LMX conducted dye, even though all the LMX vessels in this region had already developed thick secondary walls. With increased distance from the root apex, a greater proportion of the LMX vessels were stained (Fig. 7d–f) and the proportion increased to 75% at 180–200 mm and 100% at 280–300 mm. Considering the approximately four times greater diameter of LMX vessels compared with those of the EMX, the Hagen–Poiseuille equation would suggest flow rates through LMX should be around 250 times greater than in EMX. In view of this difference, once LMX became open for water transport (Fig. 7d), it required prolonged high vacuum to force dye to flow through the high-resistance pathways of EMX (Fig. 7e,f).

Summer-dormant roots As is expected, because of their mature state, EMX and LMX of dormant roots conducted dye at all distances from the apex that we could observe by the perfusion technique. It is not possible to

use this technique to assess vessel maturity very close (1–2 mm) to the apex of these roots. This, however, has been possible by measurement of ionic concentrations (see later).

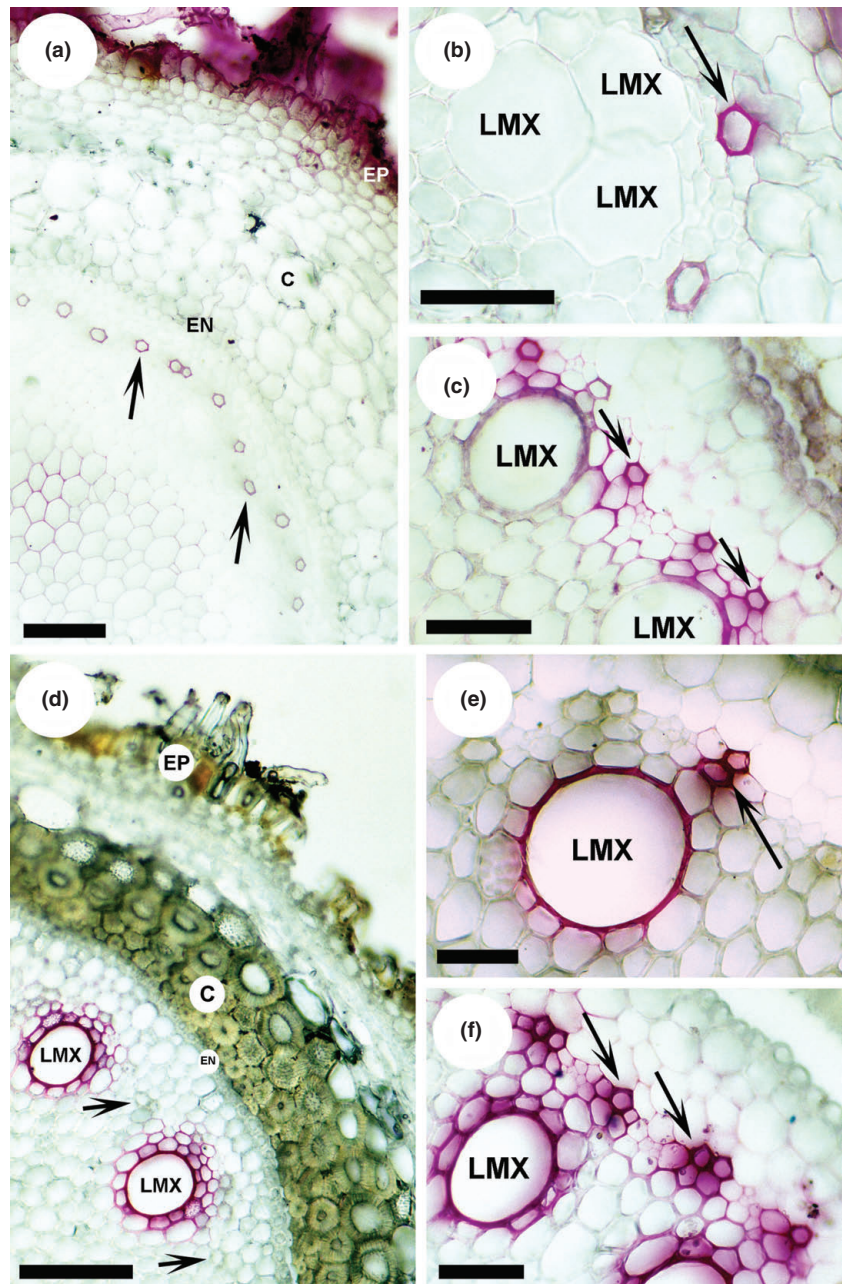
Identification of immature/mature vessel elements by microanalysis of their ionic concentrations

Xylem of winter-growing roots Potassium concentrations indicative of immature EMX vessel elements were found only in the first 10 mm from the tip. By contrast, the living LMX elements with high solutes persisted unchanged back to 100 mm and their number declined further back; all were mature beyond 160 mm (Fig. 8a). The Cl concentrations in LMX close to the root tip were lower than those of K, but Cl concentrations were similar to that of K in LMX from 50 mm back (Fig. 8b).

Xylem of summer-dormant roots The corresponding analyses of K and Cl in xylem of dormant roots showed basically that concentrations were 0 or ≤ 25 mM in LMX and EMX, indicating that all xylem was mature within 1–2 mm of the tip.

Xylem parenchyma in winter-growing roots The average K concentrations in XP (Fig. 9a) increased from 5 to

Fig. 7 Transverse sections of winter-growing roots of *Lyginia barbata* cut by hand after roots were stained with 1% basic fuchsin drawn through the xylem by reduced pressure (a–f). In (b), (c), (e) and (f), the cortex is beyond the upper right of the micrograph. In (c)–(f) dye has diffused from vessels through parenchyma walls. The green colour is likely an optical effect in some thick cell walls. (a) 10 mm from the tip; epidermal cells (EP) have produced abundant root hairs (partially obscured here by the perfusing stain, which creeps up the root surface). The thin-walled cells of the cortex (C) are intact. The walls of numerous early metaxylem (EMX) vessels (arrows) are stained. (b) Detail of (a) – dye has moved through all EMX vessels (arrow) but not through the group of three (late metaxylem, LMX) vessel elements. (c) 100 mm from the tip; walls of EMX vessels (arrows) are stained, LMX vessel elements have thick secondary walls and well defined shape, but they are still immature and do not conduct the dye. (d) 180 mm from the tip; the epidermis (EP), cells of the inner cortex (C) and endodermis (EN) have thick secondary walls. LMX vessels conduct dye solution, but EMX vessels are unstained (arrows). Dye, drawn through for ≤ 2 min, flows first through larger-diameter vessels. (e, f) 180 mm from the tip; dye solution drawn through roots for ≥ 10 min has been conducted by both LMX and EMX vessels (arrows). Bars, 100 μm (a); 50 μm (b); 50 μm (c); 95 μm (d); 27 μm (e); 35 μm (f).



10 mm from the tip, and then decreased at 50 mm; thereafter, it increased steadily further back from the tip. The Cl concentrations were, on average, lower than those of K nearest the tips, but from 50 mm back, Cl followed a similar pattern to that of K (Fig. 9b).

Xylem parenchyma in summer-dormant roots The K and Cl concentrations in XP of dormant roots increased steeply over the first 10 mm from the tip in dormant roots (Fig. 9a,b) and were of similar concentrations to XP 100–200 mm from the tip of growing roots. We did not

measure K or Cl concentrations of XP further than 10 mm back from the tip of dormant roots.

Discussion

Patterns of culm water relations and seasonal root growth

Seasonal patterns of gas exchange and water relations measurements have not been reported previously for perennial monocotyledonous species endemic to the sandplain habitat

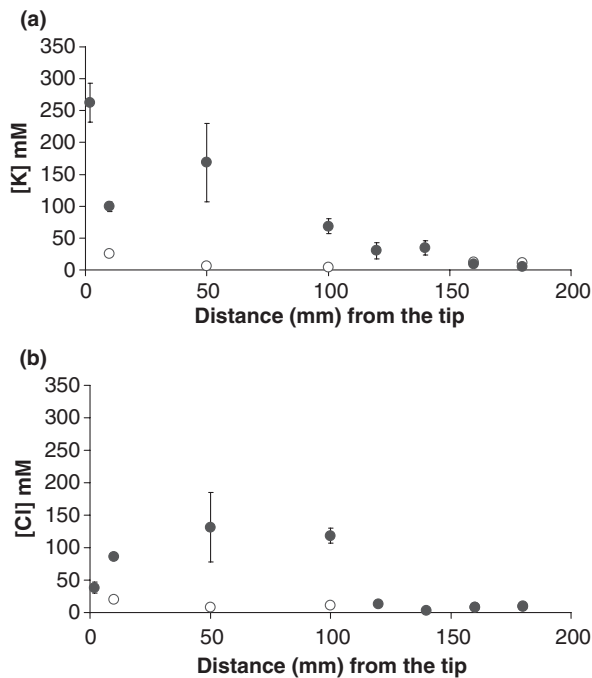


Fig. 8 Concentrations of K (a) and Cl (b) in early metaxylem (EMX, open circles) and late metaxylem (LMX, closed circles) of growing roots of *Lyginia barbata*. Each point is the average ($\pm$ SE) of 50–200 cells from 10 roots.

of southwestern Australia, but in the same habitat the water relations of several 'woody' endemics are known to be strongly influenced by seasonal variation in the moisture content of the topsoil (Dodd *et al.*, 1984). In *Lyginia*, the steady reduction in rates of photosynthesis were likely the result of decreased stomatal conductance with increasing temperatures and drying soil profiles from November through to March. The effect of drought on water relations of mature culms was relatively mild and not permanent, and culm relative water loss decreased (Fig. 3a). There was then a massive increase in photosynthesis and stomatal conductance (Fig. 3b) with the onset of cooler, wetter conditions (March to May). Clearly, water relations of the species are strongly influenced by water availability in the topsoil (to 1.5 m, Fig. 3a,b).

Water supplied by perennial deep roots (≥ 1.5 m) is a core component of the survival strategy of some perennial grasses to survive summer drought (McWilliam & Kramer, 1968) and may also represent an important trait to sustain low rates of carbon assimilation in *Lyginia* (Fig. 3b). The function of water accessed by deep roots during summer drought is associated with keeping shallow roots of *Lyginia* alive and hydrated, possibly by hydraulic lift from xylem of deep to shallow roots (Shane *et al.*, 2009). Seasonal changes in water content in upper soil layers strongly influenced the growth of shallow perennial roots (Fig. 2b–e), and resumption of growth of the previously dormant roots in wet upper soil layers in winter wet seasons (Fig. 2b–e) are clearly asso-

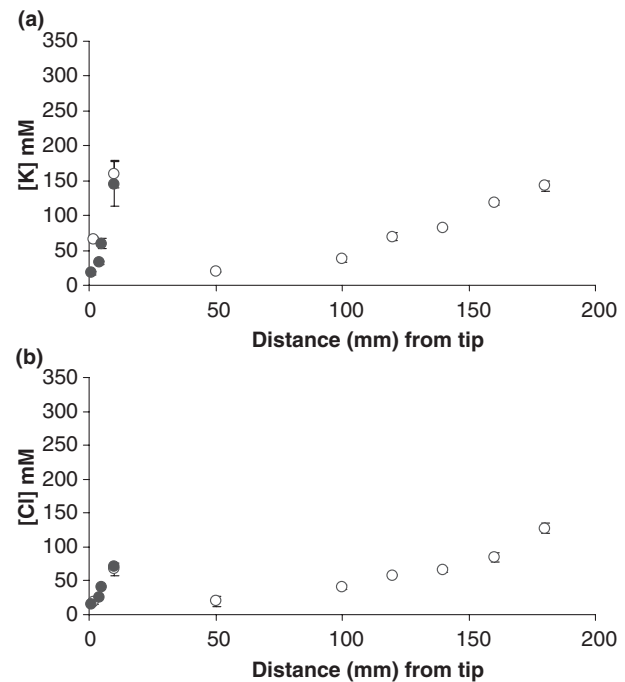


Fig. 9 Concentrations of K (a) and Cl (b) in xylem parenchyma of growing (open circles) and dormant (closed circles) roots of *Lyginia barbata*. Each point is the mean ($\pm$ SE) of 20–60 cells from 10 roots.

ciated with decreased culm relative water loss and maximal rates of carbon acquisition (Fig. 3b) and water uptake in the face of seasonal changes in soil water status (Fig. 3a).

What role(s) could root xylem–phloem juxtaposition play?

Many of the images presented in Figs 4–7 identify sites where phloem is closely juxtapositioned to EMX vessels. While higher-resolution studies were not undertaken to confirm the identity of the cells concerned, the situation in *L. barbata* may well find counterparts in branch–root junctions of maize (McCully & Mallett, 1993) and fine leaf veins (Evert *et al.*, 1978) where sieve tubes or companion cells abut directly on a vessel. Contrasting with this rarely reported anatomical arrangement, it is usual for at least one xylem–parenchyma cell to be located between conducting tissues of phloem and xylem, as suggested by Esau (1965), and elaborated upon recently in the pathways of solute retrieval suggested by Botha *et al.* (2008).

As shown in the nutrient-cycling studies by Pate *et al.* (1981) and Lambers *et al.* (1982), large proportions of the solutes (e.g. nitrogen) taken up by roots and transported to shoots in the xylem return to the root via the phloem. The significance of phloem–vessel connections in *L. barbata* roots needs to be tested, but such connections might well play a vital role in supplying metabolites to roots, as proposed for maize by McCully & Mallett (1993), and also here in summertime maintenance of water status in apices

of dormant roots by moving water with solutes acropetally to dormant apices. Previously, we found differences in osmotic potential between dormant (-1.4) and growing roots (-0.74), and significant differences in their chemical composition (Shane *et al.*, 2009). The involvement of reverse osmosis ('Münch water') in supplying water from sieve tubes for xylem transport was reviewed by Milburn (1996). Given the respiration rate of $c. 0.23 \text{ nmol CO}_2 \text{ g}^{-1} \text{ FW s}^{-1}$ determined for dormant *Lyginia* roots (table 1 in Shane *et al.*, 2009), and assuming 1 M (sucrose) in the phloem sap, it is noteworthy that in bone-dry soils dormant roots may be supplied with $c. 1.6 \mu\text{l H}_2\text{O g}^{-1} \text{ FW}$ of 'Münch water' each day (i.e. mol C respired dormant roots $\text{g}^{-1} \text{ FW d}^{-1}/12 \text{ mol l}^{-1} \text{ C}$ (sucrose) in phloem sap). If water is arriving in the dormant roots by the phloem, it seems more functional to move it from sieve tubes to parenchyma cells, and keep it away from the vessels where it would be sucked back into the shoots, at least during the day when transpiration is active. At night, downward flow of water in the mature xylem of shallow dormant roots in association with hydraulic lift from deep roots might restore water status of shallow roots. The driving force for reverse water movement could be related to differences in root osmotic pressure. We elaborate in the following sections on strategies that may help to maintain the hydration state of summer-dormant roots.

Implications of contrasting xylem maturation along shallow roots

Shallow winter-growing roots Xylem maturation, as determined by dye perfusion tests (Fig. 7) and xylem of low and high [K] (Fig. 8a), confirmed that there was a considerable distance, and hence time lag, between location of first maturation of EMX and that of LMX in growing roots of *L. barbata*. This is consistent with data for xylem maturation in many mesophytic monocotyledonous species (cf. table 1, McCully, 1995), but this report is the first to extend analysis to roots of native perennial monocotyledons adapted to a harsh Mediterranean habitat.

The [K] in vacuoles of developing LMX in *L. barbata* roots (Fig. 8a) was similar to the values measured in developing LMX of soil-grown maize (mean 110 mM, range 75–243 mM) by Enns *et al.* (1998). High [K] in vacuoles of developing xylem elements would be required for generating the turgor necessary for element expansion, but similarly high [K] in developing xylem of *L. barbata* is interesting considering the nutrient-poor conditions in its natural habitat; a general propensity for plants adapted to such habitats is to substitute Na for K; for example, see studies of xylem and phloem fluids (e.g. Banksia; Jeschke & Pate, 1995). Our CEDX studies in developing LMX typically measured [Na] at or near detection limits ($c. 25 \text{ mM}$).

In *L. barbata*, accumulation of high concentrations of Cl (Fig. 8b) in differentiating xylem might act as an osmotically balancing anion, as suggested by Huang & Van Steveninck (1988) for xylem of barley roots. In the rush, there were insufficient Cl anions to fully balance K cations in developing xylem for up to 50 mm proximal to the root tip (Fig. 8b), suggesting that other anions are involved (i.e. nitrate, malate or citrate, Shane *et al.*, 2009).

Shallow summer-dormant roots Dye perfusion tests and xylem of low and high [K] (Fig. 8a) also demonstrated clearly that differentiation of xylem reaches to within a few millimetres of the root tip of the dormant roots once its apex ceases growth and cell division. These developmental events in roots, where growth is temporarily or permanently suspended, are well illustrated in the literature (McCully, 1999; Shishkova *et al.*, 2008).

What is surprising here, however, is that open xylem would be expected to cause a significant challenge to water relations of the shallow roots of the perennial rush during the driest and hottest time of the year. The shallow dormant roots are presumably connected to transpiring culms and hence more likely to be under greater water stress compared with that of deep roots as a result of: (1) larger potential for water absorption around roots nearest to transpiring shoots (Davis, 1940); (2) significant axial resistance in deeper roots (Garrigues *et al.*, 2006; Pierret *et al.*, 2006); and (3) maturation of LMX in apical root regions of *Lyginia* dormant roots, which provides a massive increase in root hydraulic capacity, potentially resulting in $c. 250$ times greater flow. However, as stated earlier, cross-transfer of xylem-delivered water from deep roots might lead to downward redistribution of water in mature xylem of shallow dormant roots at any time when the shoot was not acting as a sink for water. The photosynthesis and stomatal conductance curves in Fig. 3b suggest that average shoot water loss is very low in the height of summer (January–March), the very conditions expected to be conducive to hydraulic lift.

It was impossible, however, using CSEM to directly observe any differences in hydration state of root cells between growing and dormant roots. There was no evidence of cell shrinkage in dormant roots whilst the plants were transpiring and xylem was not embolized (Fig. 5), in contrast to shrinkage recorded in cells of eucalypt leaves (Canny & Huang, 2006) and maize roots (Facette *et al.*, 1999) during water stress. The thick lignified walls of all cells, except phloem, that developed relatively closer to tips of summer-dormant roots could have resulted, in part, from increased soil temperatures in the sandplain habitat, because lignification of xylem has been shown to occur closer to root tips in wheat with increasing root temperature (Huang *et al.*, 1991).

The mechanisms involved in maintaining water relations of perennial roots in cacti (Ewers *et al.*, 1992) are associated with specialized xylem junctions between perennial roots and stems that act as rectifiers, allowing rapid water transport during wet conditions, but during seasonal drought, these xylem junctions embolize, inhibiting water transport from roots, and thus limit water loss. Finally, while it is tempting to speculate that 'inner cuticles' (Schneider *et al.*, 1999) and living xylem parenchyma (Fig. 9a,b), and embolism at specialized xylem junctions protect the water relations of shallow dormant roots of *Lyginia*, these are interesting areas for future investigations.

Concluding remarks

We have demonstrated a clear relationship between seasonal patterns of shoot water relations and water content in upper soil layers (≤ 1.5 m) in a perennial Southern rush. Tight control over water loss (reduced stomatal conductance) and the capacity to withstand small changes in culm relative water loss are important physiological traits for survival in extreme summer drought. However, we would view the seasonal pattern of root growth in different soil layers and the onset of root apical dormancy and eventual resumption of growth of these roots as the major driving forces of shoot water relations and growth, and for survival during drought conditions. Delayed development of LMX in growing roots and differentiation of LMX in dormant roots are important elements of this, in allowing the vascular system to transport large volumes of water collected by ephemeral branch roots borne on them at the onset of wet autumn–winter seasons.

The dormancy of only some mature roots of an individual root system is a phenomenon that may be widespread in similar habitats and needs further investigation in other species. There remains the unsolved problem of how the mature, deeply rooted *L. barbata* plants become established in the first place – that is, what features of shoot and root development allow seedlings to survive until a deep root system is established? We intend to investigate this further.

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